How Much Fault Tolerance Is in a Fly Brain?

Fault-Injection Analysis of a Whole-Connectome Spiking Controller for Physical Intelligence

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***Abstract*—Whole-brain point-neuron models built from complete wiring diagrams now serve as controllers in embodied settings, yet their fault tolerance has been probed by targeted silencing and lesioning rather than by hardware fault classes. We take the released leaky integrate-and-fire (LIF) whole-brain model of *Drosophila melanogaster* (138,639 neurons, 15.1M signed connections, 54.5M contacts) and inject five hardware-motivated fault classes—weight variability, connection loss, random-unit and hub output silencing, and sensory deafferentation—scoring the population response of descending command neurons at three to four doses, 12 seeded trials each, on the biological graph and on an edge-permuted null. Under escape-grade drive the connectome absorbs 50% lognormal weight variability; 50% connection loss retains 76.5% of command rate at a 33% latency cost; 10% random-unit output silencing leaves the median unchanged; silencing the top-1% hubs leaves the readout 26% above baseline, and silencing only their inhibitory half produces a 21× runaway that the full set masks and that costs a closed loop its steering. The null, which preserves weights and degrees but reassigns every target, never reaches threshold and decays at the same relative rate, pointing to compressive recruitment on the convergence structure rather than to redundancy alone. A synthetic closed-loop escape task then exposes the limit of these metrics: at 50% loss open-loop transmission is retained while escape collapses to 0/12, because the readout reaches its trigger only when the threat is 7–11 cm away. Five candidate design implications follow.**

***Index Terms*—neuromorphic computing, connectome, fault tolerance, fault injection, Drosophila**

I. Introduction

PHYSICAL intelligence—agents whose perception, decision, and action loops close in the physical world under tight energy and latency budgets—demands controllers that remain robust on imperfect hardware [1], [2]. Neuromorphic systems promise the required efficiency [3], and spiking implementations on Loihi 2 and FPGA substrates now cover vision and control workloads [4]–[6]. How *wiring* determines robustness remains poorly quantified: engineered networks are designed task-by-task, so their fault tolerance must be re-measured per design [7], [8]. Complete signed wiring diagrams of whole brains are public, and point-neuron models built on them reproduce realistic dynamics without training [9], [10]; an evolved connectome is thus a field-tested controller and a reference for engineered fault tolerance.

The released leaky integrate-and-fire (LIF) whole-brain model of *Drosophila melanogaster* [9][1], which instantiates the complete FlyWire connectome [10], has already been used for targeted computational silencings [9] and for network-dismantling studies of a derived sugar-sensing subnetwork [11]; we sweep five *hardware-motivated fault classes* on the fixed whole-brain model, scored jointly for transmission and activity-envelope compliance. We instrument the visuomotor escape pathway: looming-sensitive visual interneurons (LC4, LPLC2) converge onto descending command neurons (DNp01, DNa02, DNp04, DNb01) whose activation evokes take-off and escape steering [12]. Its failure modes map onto hardware fault classes: weight imprecision, broken interconnect, dead cores, dead sensors.

We inject five fault families (Section III-C), each over 12 seeded 600 ms trials, against an edge-permuted null and an analytic pathway-mass control. Our contributions: (i) a fault-injection framework mapping hardware failure modes onto a connectome-constrained 138k-neuron LIF model; and (ii) tolerance curves of descending-neuron output in a complete connectome (50% weight noise absorbed; 50% connection loss retains 76.5% of command rate, drive-invariantly; random-unit silencing leaves the median unchanged), while silencing the inhibitory half of the top 1% hubs produces a $21\times$ *over-activity* that passes transmission, violates the envelope, and costs a closed loop its steering, and deafferentation fails both criteria. We further (iii) show that graph redundancy alone explains neither specificity nor graceful degradation (compressive recruitment is a candidate account), (iv) distill five design implications, and (v) demonstrate in a synthetic closed loop that preserved transmission can still fail the task.

II. Related Work

The release of complete wiring diagrams for the adult *Drosophila* brain [10] has shifted connectome research from static graph analysis to executable models: visual-system convolutions [13], neuron-level response prediction [14], learned tasks and whole-body locomotion on whole-CNS wiring [15]–[17], biofidelic processing units [18], and descending-neuron input sharing [19], [20]. Closest to our substrate, the released LIF whole-brain model (footnote 1) reproduces global activity and has been ported to Loihi 2 [4]. Lesion studies now perturb whole connectomes—amyloid-like pathology on a fly CNS graph [21], function-preserving sparsification of the *Drosophila* model [22], and deletion of critical nodes from an SNN sugar-sensing subnetwork scored by a sensory rate threshold [11]—but none sweeps hardware-motivated fault

[1]https://github.com/philshiu/Drosophila_brain_model

classes on a fixed whole-brain controller or separates output transmission from activity-envelope compliance; we do both.

Graph-theoretic lesion studies score the structural attack tolerance of brain networks [23], not dynamics; spiking-network weight-noise robustness is mostly characterized on small trained architectures [7], [8], with adaptive mitigation now studied at the edge [24]; purpose-built fault-injection frameworks for trained SNNs—SpikeFI [25], SpikingJET [26]—automate this but score task accuracy on human-designed networks, with follow-up work on STDP-trained robustness and population-coded redundancy [27], [28]. Hardware platforms expose complementary constraints: on-chip variability co-design [5], FPGA precision–capacity trade-offs [6], permanent interconnect faults [29], and trained-CNN dependability [30]. Insect-inspired collision-aversion circuits are robust by construction but small and hand-designed [31], and fault-injection benchmarking of neuromorphic systems is only beginning [32], [33]; a complete connectome enables it at the scale of an entire evolved controller.

III. Fault-Injection Framework

A. Whole-Brain Spiking Substrate

We adopt the released whole-brain LIF model of *Drosophila* [9] (Section I), implemented in Brian2 [34] and built on the FlyWire connectome [10]. Every neuron is a current-based LIF unit (threshold -45 mV, reset -52 mV, membrane time constant 20 ms, refractory 2.2 ms, axonal delay 1.8 ms). Each connection in the released table becomes a delta-conductance synapse whose jump is 0.275 mV per counted contact and decays with a 5 ms synaptic time constant, signed by the presynaptic cell's neurotransmitter; the table yields 15,091,983 connections between 138,639 neurons (54.5M contacts), 40% inhibitory. As in the released model, baseline firing is zero and the only external input is the stimulus group, so every recorded spike traces back to the stimulated cells. Our engine restores the full network state before every trial, so conditions share initial states while the stimulus draws a fresh realization: reported dispersion includes stimulus variability. One 600 ms trial runs in $\approx$3.5 s on one CPU core.

B. Task: Visuomotor Escape

The instrumented pathway (Fig. 1) is the fly's looming-triggered escape circuit: looming-sensitive visual projection neurons LC4 (104) and LPLC2 (210) converge on descending neurons [12], [31], [35]. Because collision-sensing models treat looming as sustained high-rate drive, we drive the sensory cells (and, as a control, an equal-sized random set) with independent 150 Hz Poisson trains rather than modeling looming kinetics. The primary readout is the population rate of the command set {DNp01, DNa02, DNp04, DNb01} (8 neurons); all 1,299 descending neurons are a secondary readout. Calibration drove the command population to 94.6 Hz (interdecile 93.6–95.8 Hz, first-spike latency 4.4 ms; 10-trial set) under escape stimulation but only 8.2 Hz (6.4–9.2 Hz) under random stimulation; $\theta = 51.4$ Hz (the 94.6/8.2 Hz midpoint); every pass/fail classification below is unchanged for thresholds between the observed trial extremes (9.2–63.1 Hz).

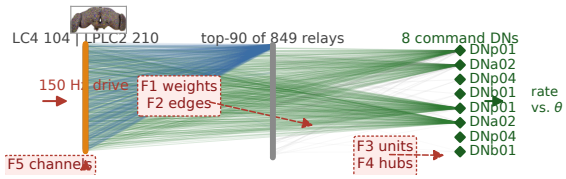


Fig. 1. The instrumented escape pathway, drawn from the connectome. Left: the 314 driven sensory cells (LC4, LPLC2); center: the 90 top-ranked of 849 two-hop relays; right: the 8 command DNs. Line width and opacity scale with synapse count (594 direct sensory→DN edges); dashed red marks the fault-injection points (F5 channels; F1/F2 synapses; F3/F4 units). Inset: reconstructed adult brain, adapted from Dorkenwald et al. [36] (CC BY 4.0).

C. Fault Taxonomy

Five fault families cover the hardware-relevant failure modes. F1 multiplies every synaptic weight by an i.i.d. mean-preserving lognormal factor ($\sigma \in \{0.05, 0.1, 0.2, 0.5\}$), modeling analog weight variability and bit-level imprecision. F2 deletes a random fraction $f \in \{0.05, 0.10, 0.25, 0.50\}$ of connections ($\approx f$ of contacts in expectation), modeling interconnect or memristor failures. F3 silences all outgoing connections of a random fraction $f \in \{0.01, 0.02, 0.05, 0.10\}$ of units, excluding input and readout cells (silenced cells still receive input and spike), modeling dead cores at the output boundary. F4 applies the same operation to the top 1%, 2%, or 5% of units ranked by total outgoing contact count, modeling failure of high-fanout components; these cells carry 9–22% of all connections, a targeted, correlated version of F3. F5 zeroes all outgoing connections of one (LC4 or LPLC2) or both sensory channels, modeling sensor or afferent-path failure. All faults are applied after state restoration; F4 and F5 act on fixed cell sets, while F1–F3 redraw their realization each trial from SHA-256 seeds over (phase, condition, trial)—12 independent draws per condition; fault masks are bit-reproducible, whereas the stimulus is not re-seeded, so whole re-runs are statistically equivalent rather than bit-identical.[2]

D. Controls

Two controls isolate what the biological wiring contributes. The first is an edge-permuted null: every synapse keeps its presynaptic partner, weight, and sign, but its postsynaptic target is randomly reassigned. This preserves out-degrees, weights, and the global in-degree histogram while destroying which inputs converge onto each target—and with them all multi-hop structure; the identical sweep runs on it. The second is analytic: for both graphs we compute the *pathway mass* $m_k = \mathbf{1}_{\mathrm{DN}}^{\top} A^k \mathbf{1}_{\mathrm{in}}$, the synapse-weighted total of all k-hop directed paths from LC4∪LPLC2 into the full descending layer (m_k sums k-step walks of either sign—a density proxy, not signal flow), on the intact graph and after the F2 deletion levels; comparing its decay with the output decay separates topological redundancy from dynamical amplification.

[2]Code, connectome-derived edge lists, and per-trial records: https://github.com/Scophield/drosophila-fault-injection.

IV. Results

A. Baseline: The Connectome Is a Specific, Low-Latency Controller

Driving LC4 and LPLC2 at 150 Hz produces a command rate of 94.7 Hz (median over 12 trials) with 4.05 ms first-spike latency, far above θ. An equal-sized random set, driven identically, yields only 8.2 Hz on the same readout while producing *more* total activity (293k vs. 40k spikes), so the readout is pathway-specific rather than a generic arousal response. With both channels silenced ($F5_{both}$) it collapses to zero in every trial, and the residual 21.2k spikes are the stimulated sensory cells themselves, so the score measures transmission through the pathway.

B. Tolerance Curves: Noise Is Absorbed, Disconnection Degrades Gracefully

Figure 2 shows dose–response curves for the two most consequential families; F1 and F3 are flat (Section V). At σ=0.5 weight variability leaves the median unchanged (94.2 vs. 94.7 Hz) but widens trial dispersion (interdecile 90.4–100.8 vs. 94.0–95.6 Hz) and raises total activity only from 39.7k to 41.6k spikes. Random connection deletion degrades gracefully: 5% and 10% loss leave 91.9 and 93.1 Hz, 25% retains 84.5 Hz (89.2%), and 50% retains 72.4 Hz (76.5%), with all 12 trials passing transmission in every condition (weakest 63.1 and 68.1 Hz, $\geq$22% above θ). The cost appears predominantly in latency—4.05 to 5.4 ms at 50% loss (+33%)—consistent with weakened aggregate drive rather than longer residual routes; Silencing 10% of units at random—nearly 14,000 neurons cut off from their targets—also leaves the median unchanged (95.1 Hz): command drive integrates over hundreds of converging populations, and no tested random slice is load-bearing. The 1,299-neuron descending layer tracks the same pattern (2,941 vs. 1,918 spikes at 50%), so the graded response belongs to command encoding, not to the eight-neuron readout.

C. Hub Silencing Is Resolved by Sign, Not by Size

The dose–response of hub silencing (F4) is the only non-monotonic family (Figs. 2b, 3b). Silencing the top 1% of cells by outgoing contact count (1,386 neurons, 9.1% of connections, 62% of them inhibitory) raises the command rate to 119.1 Hz (+26%) and global activity to 49.3k spikes (1.25× healthy; Mann–Whitney $p < 10^{-4}$); at 2% (13.2% of connections) it stays elevated (114.8 Hz, 38.9k) and at 5% returns to baseline (94.0 Hz, 34.1k). Splitting by sign explains both: silencing only the 765 inhibitory hubs (5.6% of connections) drives the readout to 168.2 Hz and activity to 846.9k spikes (21.4×) in every trial, whereas silencing only the 621 excitatory hubs (3.5%) lowers them to 88.6 Hz and 32.3k (0.82×). The full set thus nearly cancels a runaway against a depression, which is why aggregate removal looks mild while its inhibitory half is extreme. Identity, not mass, drives this: 765 random inhibitory cells remove 0.8% of connections and change nothing (95.2 Hz), and even 8,889 random inhibitory cells removing 8.6%—more than the hub set—give only a 1.27× median excursion (102.4 Hz, 50.3k), though one of twelve trials runs to 327k. A set matched to the hubs in size and E/I composition (94.5 Hz, 40.1k, 1.1% of connections) and random 10% unit silencing (95.1 Hz) likewise do nothing. Scoring *transmission* (rate $\geq \theta$) and an *activity envelope* (within 2× healthy) separately, the inhibitory-hub condition alone passes the first and violates the second in all 12 trials; every distributed fault passes both, and $F5_{both}$ fails both.

D. What the Wiring Contributes: Null Network and Pathway Analysis

The permuted null dissociates specificity from redundancy: out-degrees, weights and the in-degree histogram survive but convergence is reassigned, and no trial reaches θ (median 0.0 Hz). Its readout falls to background (45.5 vs. 2,941 spikes per trial) and global activity to 60% (23.8k vs. 39.7k), barely above the 21.2k the driven cells produce with all outputs severed. Pathway mass quantifies the convergence advantage: 5.8× (1-hop) and 17.7× (4-hop) more route mass into the descending layer. Yet *relative* pathway-mass decay is nearly identical (at f=0.5, k=8: 0.3% vs. 0.4% of baseline; Fig. 3a), while output degrades far more slowly (76.5% at f=0.5 against 5.8% four-hop route mass), ruling out graph redundancy alone. A candidate account is compressive recruitment: with N converging afferents and survival p, aggregate drive scales as Np while k-hop route mass decays as p^k (6.25% at k=4, p=0.5), and a threshold followed by saturation compresses drive changes into sub-proportional rate changes. Retention exceeds proportional scaling at every dose and increasingly so as damage weakens (1.5× proportional at p=0.5, 3.6× at p=0.25, 19× at p=0.05): mild deletion is buffered almost entirely, severe deletion bites. At p=0.5 the buffering is drive-invariant—raising drive from 75 to 300 Hz lifts the healthy rate from 75.2 to 113.0 Hz but leaves retention at 73.8%, 77.6%, 76.5%—so it is set by convergence geometry rather than by operating point, although a common saturated regime at all three drives cannot be excluded. A single power law fits only the severe end ($\alpha \approx 0.4$), so we report compression as dose-dependent rather than as a global exponent.

E. From Transmission to Behavior: A Closed-Loop Escape Demo

Transmission retention need not imply task capability, so we close the loop: looming rates over the 314 LC4/LPLC2 cells (assumed receptive fields) drive the full network, a frozen left–right difference across the eight command cells steers a planar agent away from a pursuing threat, and takeoff fires when their mean rate crosses θ_{demo}=25.6 Hz (median peak of six training episodes; the open-loop θ=51.4 Hz does not transfer). It escapes 24/24 across four azimuths ($\pm30°$, $\pm60°$; six seeds; median miss 0.09–0.11 m against a 0.05 m radius; Fig. 4), including 12/12 at the untrained azimuths ($+30°$, $+60°$), and 0/3 for a frontal approach (blind spot). Every control collapses (0/12 each), while a scene-reading hand rule reaches 12/12. Faults transfer as a threshold cliff rather than a graded rate loss: 25% loss escapes 10/12 and 50% escapes 0/12, although

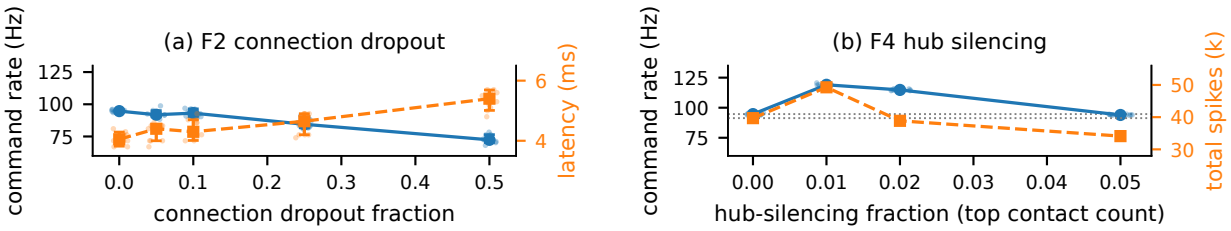


Fig. 2. Tolerance curves of descending-neuron output (medians over 12 trials; blue: rate; orange: latency (a) / total spikes (b); dotted grey: healthy medians). (a) Connection dropout: 76.5% of command rate retained at 50% loss, latency +33%. (b) Hub silencing: non-monotonic dose response.

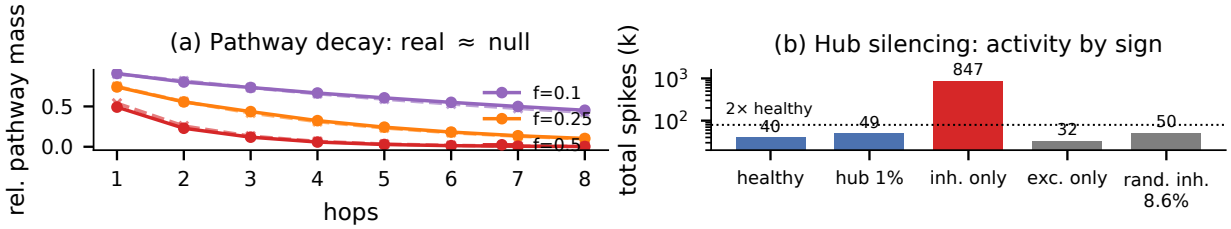


Fig. 3. Topology contrast. (a) Route mass under edge loss, normalized per graph (connectome solid, null dashed): relative decay nearly identical, absolute mass 5.8–17.7× higher. (b) Total activity by sign (medians over 12 trials, log scale; dotted: 2× healthy envelope): the inhibitory half runs away, the excitatory half is depressed, and a random inhibitory lesion of 8.6% of connections does not. Null readout 0.0 Hz in every condition.

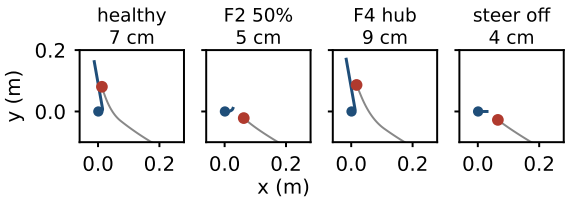


Fig. 4. Synthetic closed-loop escape at $-30°$ (seed 0, one threat script). Blue: agent path; grey: threat path; red dot: threat endpoint. Columns: healthy, $F2_{50}$, $F4_{1\%}$, steering disabled; the distance in each label is the closest approach. Success: separation > 0.05 m.

the same 50% loss retains 72.4 Hz open-loop. Under single-target drive the $F2_{50}$ readout peaks at ≈38 Hz and first reaches θ_{demo} only at 420–500 ms, when the threat is 7–11 cm away—too late for the 250 ms aiming delay and 450 ms takeoff—so preserved transmission does not imply preserved behavior and the open-loop criterion is the wrong certificate for a closed loop. Hub silencing, by contrast, preserves escape (12/12) at 1.7× the loop's healthy activity (1.25× open-loop), inside the 2× envelope. Its inhibitory half is the dangerous mode: 6/12 escapes, 1.24M spikes (77×), takeoff firing in every trial while the median heading change falls from 97° to 64°, and all six failures at the two trained azimuths—the runaway does not silence the controller, it destroys the steering signal. This is a preliminary demonstration, not a behavioral validation (assumed encoding, scripted envelope, kinematic agent).

V. Candidate Design Implications for Embodied Neuromorphic Controllers

Three of the five implications concern where redundancy pays. Because 50% weight variability and 10% unit-output silencing leave the median command rate unchanged (99.4% and 100.4% of baseline), precision and per-unit yield can be traded against wiring redundancy: weights may drift by a lognormal σ of 0.5—beyond most analog or multi-bit resistive devices—and 10% of cores may fail outright, provided many routes converge [5], [7]. Disconnection is paid in latency, not rate—76.5% of the rate survives at 50% loss while latency rises 33%—so a real-time loop should budget that curve (Fig. 2a).

The other two implications concern failure modes that per-unit monitoring misses. Hub silencing is the only class that can *amplify*, and its sign matters more than its size: removing the inhibitory half of the top 1% raises activity 21×, while the full set hides this behind a matching excitatory loss—so monitor the activity envelope, not liveness; and since 8.6% of inhibitory mass removed at random was benign in eleven of twelve trials while 5.6% removed at hubs always destabilised, sparing high-fanout *inhibitory* units is the cheaper mitigation. Deafferentation is the opposite extreme: either channel alone retains 68–79% of the median rate, but losing both drives the pathway to zero, making the sensor interface—not the synaptic fabric—the first redundancy candidate. Because relative pathway decay is identical while only the biological graph yields a decodable command, fault injection belongs pre-deployment [32], [33].

VI. Limitations and Conclusion

Our LIF abstraction fixes one drive rate, the hub sign split is measured at that one rate, and correlated damage is excluded. One pathway is instrumented, so the ordering is a hypothesis for others rather than a whole-brain law, and its vulnerability to aging [37] suggests tolerance is not fixed either. Under the tested drive the median rate survives weight noise and unit silencing, retaining 76.5% at 50% loss; the dangerous failure is inhibitory-hub over-activity, which a closed loop turns into lost steering.

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